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import tkinter as tk
from tkinter import messagebox
import random

class RobotLocalization:

    def __init__(self, root):
        self.root = root
        self.root.title("Mobile Robot Localization System")
        self.root.geometry("1000x700")

        self.x = 100
        self.y = 100
        self.radius = 15

    def move(self, dx, dy):

        new_x = self.x + dx
        new_y = self.y + dy

        if self.check_collision(new_x, new_y):
            messagebox.showwarning(
                "Warning",
                "Robot cannot move through obstacle!"
            )
            return

        self.x = new_x
        self.y = new_y
        self.update_robot()

    def check_collision(self, x, y):

        for obstacle in self.obstacles:
            if (
                x > obstacle[0] and
                x < obstacle[2] and
                y > obstacle[1] and
                y < obstacle[3]
            ):
                return True

        return False

    def update_robot(self):

        self.canvas.coords(
            self.robot,
            self.x - self.radius,
            self.y - self.radius,
            self.x + self.radius,
            self.y + self.radius
        )

        self.position_label.config(
            text=f"Position: ({self.x}, {self.y})"
        )

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